

YUECONG MAO

Target: Fall 2028 PhD / Available for Research Internships Now

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EDUCATION

Beijing Institute of Technology
B.Eng. in Artificial Intelligence

Beijing, China
Expected June 2028

Relevant Coursework: Programming Methodology and Practice (100) | Literature Retrieval and Academic Integrity (95) | Database Principles (91) | Introduction to Cognitive Science (90)

RESEARCH INTERESTS

I focus on **principled and efficient LLM training**, with an interest in **data selection and training-process optimization**. By studying how data value and optimization strategies evolve across different stages of training, I aim to develop methods for more effective data selection and better trade-offs between training cost and model performance.

Research Question: *What data is most valuable to a model at each stage of training, and how can the training process be optimized to make the best use of it?*

RESEARCH EXPERIENCE

Visual Evidence-based Reasoning for Multi-Object Multi-State Transitions

W. Zhang, Q. Zhang, **Y. Mao**, ..., X. Wu*

ICLR 2027 (Under Review)

Personal Contributions:

- Contributed to a **multi-object video dataset** by curating 800+ YouTube videos, reviewing 15,000+ clips for temporal state transitions, and analyzing annotation errors to guide targeted prompt modifications for Qwen-based models.
- Hypothesized that speech could resolve **visual ambiguity** in object-state annotation (e.g., salt vs. sugar); introduced a **speech-aware annotation pipeline** using Qwen3-ASR-1.7B and Qwen3-ForcedAligner-0.6B to extract and align spoken content, achieving 95% object accuracy across 534 videos and improving annotation pass rate over the original visual-only setting.
- During **comparative experiments** with Video-R1-7B and Qwen2.5-VL-7B, optimized long-video inference under limited GPU memory through 12-second segmentation, 8 FPS sampling, and reduced spatial resolution.
- Uncovered temporal misalignment between predictions and ground truth; designed a **hierarchical matching module to align video segments** before evaluation, increasing physical event recall by 34.3 percentage points.

MATAR: Multi-Agent Task-Adaptive Refinement for Multimodal Data

Y. Ai, X. Li, **Y. Mao**, ..., R. Li*, G. Wang*

ICLR 2027 (Under Review)

Personal Contributions:

- Designed a **Paired Replay Evaluation** (identical splits, model architecture, training steps, and locked seeds; only the data differs) to disentangle true refinement gains from training variance, providing a **high-confidence unbiased benchmark** for 9 SOTA data-selection baselines.
- Designed a **multimodal corruption pipeline** to systematically perturb text-image features and labels in 10% increments; found that **MATAR remained robust under up to 90% modality corruption**, outperforming baselines by 1.56% and 2.68%, but lost its advantage under severe label noise, where accuracy converged to 14.3%, identifying a clear **label-noise failure boundary**.

- Demonstrated **zero-shot transfer** across 8 domain pairs and 4 task types, with consistent downstream gains (e.g., +2.80% Macro-F1 on Grocery→Movies and +1.36% MRR on Bili Music→Flickr30k); identified **task-specific refinement patterns**, showing that classification benefits from graph-topology rules while retrieval/matching rely more on feature alignment and semantic-context reconstruction.

OpenMAS-GAPP: Graph-Guided Construction and Evaluation of Executable Multi-Agent Reasoning Processes

CVPR 2027 (Targeted)

X. Li, K. Yang, J. Li, Y. Mao, ..., R. Li, G. Wang**

Personal Contributions:

- Ran **GraphHarness end-to-end** on four existing domain-specific QA benchmarks, configuring datasets and troubleshooting runtime issues to validate the four-stage Harness construction pipeline—from input, requirement model, blueprint, to Executable MAS.
- Benchmarked GraphHarness against 10 baselines through **token-accuracy analysis**, identifying inefficient high-token reasoning regimes despite limited accuracy gains.
- Built an **error-driven refinement pipeline** by extracting and analyzing incorrect cases individually. Identified a failure mode where the adapter degraded rich natural-language requirements into oversimplified JSON, causing missing required fields in executable MAS construction; introduced a **natural-language-to-JSON encoder** to preserve task information, correcting 48.3% of previously incorrect cases, while only 5.1% of previously correct cases changed outcomes.

SELECTED PROJECTS

Transformer-based Sentiment Prediction Programme

NUS School of Computing — Road to ChatGPT

- Conducted controlled experiments on a Transformer-based sentiment classifier, varying key hyperparameters including learning rate, embedding dimension, and regularization.
- Compared model performance across configurations and analyzed the effects of different model settings.

LittlePopPulse — Labubu Humanoid Robot Simulation Contributed to Blender-based motion visualization and implemented 10 predefined robot behaviors using URDF. [\[GitHub\]](#)

TECHNICAL SKILLS & LANGUAGES

- **Programming Languages:** Python, C/C++, SQL
- **Frameworks & Tools:** PyTorch, Ubuntu
- **Languages:** IELTS 7.5 overall

HONORS & AWARDS

- **Outstanding Student Leader Award** | 2025
- **Class President (University Outstanding Class)** | 2025
- **Second-Class Undergraduate Academic Scholarship** | 2024, 2025
- **Second Prize, National College Student Mathematics Competition (Beijing)** | 2025
- **Third Prize, 9th National College Student Integrated Circuit Innovation and Entrepreneurship Competition (North China Division)** | 2025